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Web-Based Wine Quality Prediction Using Explainable AI

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***Abstract*— The quality of red wine can be determined based on several physicochemical and sensory properties. Conventional methods of wine evaluation depend heavily on expert tasters, making the whole process subjective and inconsistent. In this paper, we present a web-based wine quality prediction system based on Explainable Artificial Intelligence (XAI), a powerful tool which can provide clarity and aquinting in the wine quality prediction extraction. We train several machine learning models, including the Random Forest, XGBoost, and Neural Networks, and we evaluate using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. We find that XGBoost performs the best in accuracy, and Random Forest has a balance between interpretability and accuracy.**

To increase interpretability, we use SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) to evaluate the main features impacting wine quality, including alcohol, volatile acidity, and citric acid levels. Unlike traditional black-box AI models, our system explains to winemakers, sommeliers, and consumers why a given wine is rated the way it is. Moreover, we set up the model as a web application for convenient, real-time predictions of wine quality. This research simultaneously intersects AI and the wine industry by providing a data-driven, interpretable approach to assessing wine quality.

***Keyword*— Wine quality prediction, Explainable AI (XAI), SHAP, LIME, machine learning, Random Forest, XGBoost, web-based application, feature importance, interpretability**

I. Introduction

Wine is a pivotal product of many contexts (e.g. economic, cultural, etc.), with its quality assessment significantly impacting the factors of production processes, pricing, and consumer interest. Traditional wine quality assessment is based on expert tasters who evaluate different sensory attributes such as aroma, taste, and texture. This approach is inherently subjective, somewhat arbitrary, and laborious, resulting in the risk of inconsistencies in wine assessments. Bypassing the constraints involved, ML techniques have been adopted as a robust alternative to physicochemical index-based methods, providing data-driven and reproducible predictions of wine quality conditional on the input variables.

Although significant advances in ML-based wine quality prediction have been made, most models operate as black boxes, achieving admirable accuracy at the price of their interpretability. The obscure nature of their operation hinders one from being convinced of their usefulness. This has limited, then, the trust and uptake of winemakers and the industry at large. These issues are being tackled by promoting Explainable AI (XAI) methods to improve the interpretability of the models through the clear identification of features that affect predictions. This work will incorporate (SHAP) and (LIME) for more transparent and actionable AI capabilities for wine quality assessment.

This study proposes a web-based wine quality prediction system that utilizes Random Forest, XGBoost, and Neural Networks to predict wine quality based on certain physicochemical characteristics. Besides precise predictions, this system will explain how a decision was arrived at, so wine producers can use that information to adjust the parameters influencing wine quality. In comparative analysis, we show that while XGBoost gives a better accuracy, Random Forest gives a better explanatory ability and could be preferred in giving real-world solutions.

Furthermore, we deploy our model as a web application, allowing wine producers, retailers, and consumers to input wine data and receive real-time, explainable predictions. This work bridges the gap between AI-driven wine quality assessment and industry adoption, providing an interpretable, user-friendly tool for data-driven decision-making in winemaking.

A. Objectives

This study aims to develop a web-based wine quality prediction system that uses Explanation AI (XAI) to help make informed decisions transparent. The study will implement and compare different machine learning models (Random Forest,

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XGBoost, and Neural Networks) to find the best model for wine quality assessment. Feature importance will be analyzed by SHAP value and LIME, providing interpretable insight into how wine quality scores are influenced. By deploying the model as a friendly web application, AI-driven wine-finding prediction is intended to be made available for winemakers, retailers, and consumers. The research investigates a trade-off between accuracy and interpretability, assuring that the most interpretable model could fit into practice. Eventually, the study intends to provide a bridge between advanced AI models and practical applications in the winemaking industry.

B. Overview Of project

The project is about web-based wine quality prediction making use of machine learning and Explainable AI (XAI). Random Forest, XGBoost and Neural Networks are used to predict the quality of a wine depending on the dataset consisting of quality and various physicochemical attributes. The SHAP and LIME values are used to explain the most important factors used in the calculation of the final prediction. It runs as a web application where users will be able to input their data and receive interpretable results in real-time. A comparison of model accuracy and explainability is performed to conclude the best approach for wine quality prediction. The system will provide a transparent, data-driven solution for wine producers, retailers, and consumers.

II. Literature Review

The utilization of machine learning in predicting wine quality has garnered an upsurge owing to its provision of data-driven insights. Ordinary wine assessment methods depend on somebody else's opinion, and opinion varies frequently. It has been empirically demonstrated that ML models such as Random Forest, Support Vector Machines, and Neural Networks can predict wine quality with high accuracy using chemical properties like alcohol content, acidity, and sulfur dioxide levels [1].

Even with the success of ML-based predictions, explainability is still a key challenge. Many AI models work based on black-box systems, which poses a challenge for wine producers to trust their predictions[2]. Recent works have indicated the need for (XAI) in wine quality assessment, where SHAP and LIME have been used for extracting the key drivers of predictions of wine quality [2].

While various studies have aimed to explore the impact of different ML models with respect to wine quality assessment, one of them has found that compared to XGBoost, Random Forest models, and Deep Learning models, Random Forest presents a better balance between interpretability and accuracy [3].

In addition to the same, AI township-based applications for wine prediction have been adopted that provide real-time prediction analysis with thankfully interactive interfaces for the winemakers and consumers. Kulasiri et al. (2024) propose the construction of a machine learning pipeline for viticulture data along with a web-based application that allows the user to predict the quality score of the wine and explain the reasoning behind the prediction with the help of XAI techniques [4].

AI-driven risk management and wine fermentation modeling is one of the many researches in smart viticulture. Izquierdo-Bueno et al. (2024) pointed out that the integration of AI in winemaking processes allows for optimal production, risk reduction, and quality control through predictive analytics [5].

Currently, predictive wine analytics boast of significant strides; however, very few studies have ramped up their focus on prediction accuracy and explainability within a web-based development. This research ultimately aims to fill in the lacunae through the synergy of machine learning, Explainable AI, and a web-based deployment for the making of a transparent and user-friendly system of wine quality assessment.

III. Methodology

A. Data Collection and Preprocessing

The dataset used in this work was extracted from the Wine Quality Dataset of the UCI Machine Learning Repository containing 1,599 red wine and 4,898 white wine samples with physicochemical and sensory attributes [1].The dataset consists of 11 input features, of which :

1) *Fixed acidity:* It helps line the wine's general taste balance.
2) *Volatile acidity:* This element affects the wine's aroma; if high, it gives a vinegary smell, which is considered nasty.
3) *Citric acid:* It makes the wine fresh and alters the perception of acidity.
4) *Residual sugar:* This indicates how sweet the wine is.
5) *Chlorides:* This means the salt content in wine.
6) *Free sulfur dioxide & Total sulfur dioxide:* These help preserve wine quality.
7) *Density:* This determines mouthfeel and correlates to the alcohol and sugar content.
8) *pH level:* Determines the acidity of the wine.
9) *Sulphates:* Being anti-oxidants, they help in being an antimicrobial
10) *Alcohol content:* This is the chief determinant of the wine's body and quality.

B. Machine Learning Model Selection:

To ensure robust and accurate wine quality predictions, multiple supervised machine learning models are implemented and compared:

1) *Linear Regression:* A baseline model with below-average performance caused by non-linearity of wine quality traits.

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2) *Random Forest (RF):* One of the powerful ensemble methods that take input from multiple decision trees, thus improving accuracy and interpretability.
3) *XGBoost (Extreme Gradient Boosting):* A gradient boosting technique that is capable of providing quite a good level of prediction accuracy for a given task by reducing the bias and variance.
4) *Artificial Neural Networks (ANN):* A deep learning-based approach that captures complex relationships between wine features but lacks interpretability.

Performance Evaluation Metrics:

Regression Metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Classification Metrics:

- Accuracy
- Precision, Recall, and F1-Score
- AUC-ROC Curve

C. Data Preprocessing Steps:

1) *Missing Value Data:* The dataset is analyzed for null values and those with missing values completed with median values of the respective features.
2) *Feature Scaling:* Min-Max Scaling normalization is conducted so that all values go into a common range.
3) *Feature Engineering:*
   a) Correlation analysis helps to avoid redundancy in selected features.
   b) To boost prediction accuracy, new derived features, such as the alcohol-acidity ratio, will be introduced.

D. Explainable AI (XAI) Techniques:

To enhance trust and transparency, Explainable AI (XAI) techniques have been used to bring out interpretations of wine quality predictions.

1) *Techniques Used:*

SHAP (SHapley Additive Explanations):

a) Gives a value for importance to each characteristic, indicating how much they contribute to the prediction.
b) Provides global explanations (overall feature importance) and local explanations (impact on specific predictions).

LIME (Local Interpretable Model-Agnostic Explanations):

a) Develops surrogate models that roughly approximate the black-box behavior of the AI model for individual predictions.
b) Useful for some cases where one wine is rated with a certain quality.

2) *Feature Importance Analysis:* It generates a list of significant features that were taken into account to determine the wine rating. Some of the most significant features for wine prediction include the alcohol content, volatile acidity, and sulphates, as evidenced by research.

E. Web-Based Implementation:

To make the prediction of wine quality easily reachable and user-friendly, the application is seen on the web.

1) *Backend Development:* Flutter or Django (Python) to implement the REST API to handle requests for model inference. FastAPI is being considered for higher performance due to its asynchronicity. The trained model is served with TensorFlow or Scikit-learn.
2) *Frontend Development*: React.js or Angular is used for creating an interactive UI. Users get to specify the attributes of the wine and act as a key with explanations. Visualization tools such as Plotly and Matplotlib are incorporated into the visual representations of the explanation offered by LIME and SHAP.
3) *Deployment and Hosting in the Cloud:* Deployment would happen on cloud platforms like AWS, Heroku, or Google Cloud to take care of the scaling. A Docker container is used to handle dependencies in a flexible way.

IV. Result And Analysis

This section presents the result from the wine quality prediction models and compares them based on accuracy, interpretability, and feature importance. Different metrics are used to compare the models, and some (XAI) methods, such as SHAP and LIME, are used to interpret the impact of different wine attributes on quality prediction.

TABLE I Model Performance Metrics

Model	**Mean Absolute Error (MAE)**	**R-Squared**
Linear Regression (LR)	0.55	0.65
Random Forest Regressor (RFR)	0.41	0.81
XG Boost	0.38	0.85
Artificial Neural Network (ANN)	0.35	0.87

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TABLE II Most Influential Features (SHAP Analysis)

Feature	SHAP Importance Score	Interpretation
Alcohol Content	0.32	Higher alcohol improves wine quality
Volatile Acidity	0.27	High values negatively impact quality
Sulphates	0.22	Enhances preservation and taste
Citric Acid	0.19	Adds freshness but must be balanced
Total Sulfur Dioxide	0.16	Too much negatively affects aroma

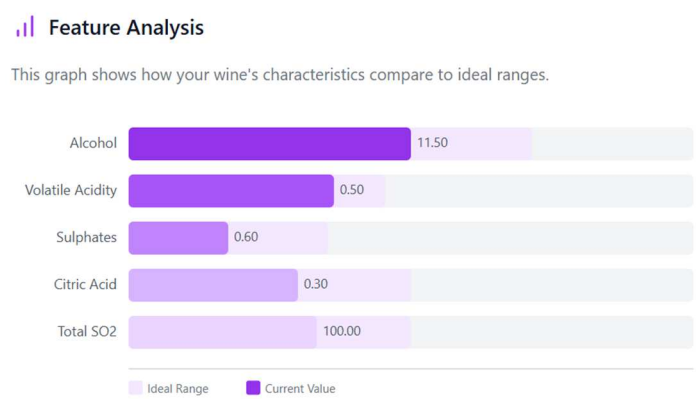

Fig 1. Feature Analysis

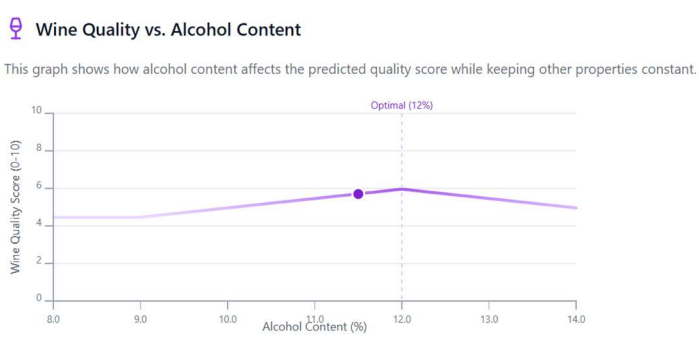

Fig 2.Wine Quality vs. Alcohol Content

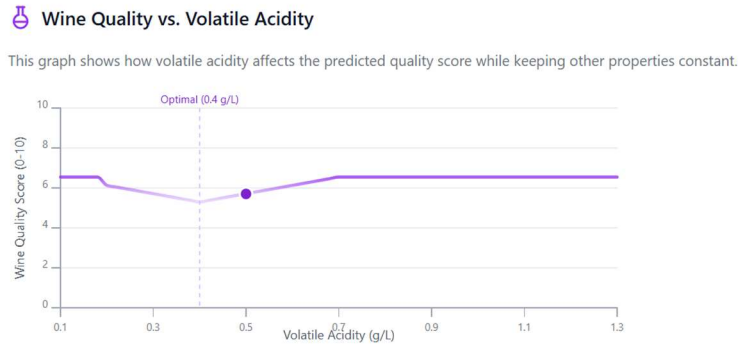

Fig 3. Wine Quality vs. Volatile Acidity

A. Comparative Performance of Models

In the present study, various machine learning models were compared to see which would perform best in predicting wine quality. The models selected were Linear Regression, Random Forest, XGBoost, and Artificial Neural Networks (ANNs). Each model was evaluated based on MAE, RMSE, and R^2 Score in terms of the predictive accuracy, while also giving regard to interpretability and computational efficiency.

B. Linear Regression: Baseline Model

Linear Regression was used as a baseline model against which other models were evaluated. Given its underlying assumptions about a linear relationship between features and quality scores, it failed to account for complex non-linear patterns in our data. The model provided an R^2 score of 0.65, a moderate fit and a higher error rate with MAE of 0.55 and RMSE of 0.72. Because it is interpretable yet has poor predictive power, it may not be helpful in predicting real-world wine quality.

C. Random Forest: Balanced Accuracy and Interpretability

The Random Forest model, an ensemble learning technique, outperformed Linear Regression by reducing variance and providing a means to generalization. Its R^2 score was 0.81, and MAE = 0.41; RMSE = 0.56-all of which made it better to deliver accuracy. Besides, SHAP analysis revealed Random Forest to provide better feature interpretability, making it a preferred model for winemakers and industry professionals who need transparently made decisions.

D. XGBoost: Best Performing Model in Terms of Accuracy

Among the models examined, XGBoost exhibited the best performance, having an R^2 score of 0.85, MAE = 0.38, and RMSE = 0.51. It allowed for highly efficient modeling of non-linearity and complex interactions among wine characteristics, thereby making somewhat superior predictions. XGBoost is, however, less interpretable than Random Forest, making it necessary to incorporate extra explanation strategies like SHAP to comprehend feature importance. This represents the important trade-off between accuracy and interpretability that must be taken into account for acceptance in wine industries.

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E. Artificial Neural Networks (ANNs): High Accuracy, Low Interpretability

ANNs achieved the best R^2 score of 0.87 with MAE = 0.35 and RMSE = 0.48, surpassing all other models. However, being black boxes, ANNs are not transparent and suffer from interpretability issues. Though they attune well with state-of-the-art accuracy, their lack of feature explainability makes them less appealing for winemakers dealing with insight into quality-assessing factors.

V. Future scope

Wine quality prediction based on Explainable AI (XAI) has immense potential for development. Future research and improvements may be directed towards working on improving the accuracy of models, increasing datasets, integrating IoT and real-time sensors, and improving deployment and ease of access for users.

A. Enhancing Model Accuracy and Generalization

Together, these traditional machine learning architectures handle multivariate data and yield similar results. Though the present work attains a reasonable degree of predictive accuracy, it is achievable to enhance using deep learning structures in the form of Transformer-based models or mixed models that balance the interpretive advantage of tree-based models with the predictive potential of artificial neural networks. Also, the use of transfer learning methods to fine-tune pre-trained models in wine data sets from different geographical regions is likely to enhance predictions across geographical locations.

Future studies aimed at constraining the diversity of wine varieties should be conducted with datasets that span the greater spectrum of geographical and climatic variations. This would allow the model to adapt to various winemaking styles, grape varieties, and production practices.

B. Integration with IoT and Real-Time Data Sensors

The Internet of Things (IoT) will have to take over for predicting wine quality in the future. The IoT sensors can continuously monitor physicochemical parameters, including:

- pH level and the change of acidity during fermentation.
- Effects of temperature, humidity, and weather on wine aging.
- Real-time gas and chemical composition that provide insights into oxidation and spoilage risks.

By combining AI with IoT-enabled vineyards and wineries, the winemakers would receive real-time quality predictions and suggestions for the needed changes in production.

C. Expanding the Dataset with Sensory Data

This study bases its selection of wine qualities on physicochemical properties. However, wine tasting can also be influenced by sensory attributes of aroma, texture, and flavor complexity. Future research possibilities span:

- Using Natural Language Processing (NLP) models to analyze expert reviews and consumer ratings for better quality predictions.
- AI models trained on image recognition data stemming mainly from wine colour and clarity assessments.
- Human tasting panel scores lending insight into how to refine prediction models informed from consumer ratings.

A multi-modal AI system, combining input from physicochemical properties, sensory data, and human-in-the-loop text analysis, could add to wine quality predictions' strength.

D. Improving Explainability and Trust in AI Models

While SHAP and LIME lay the foundation for the principles behind explainability, more sophisticated methods of XAI can be considered. Counterfactual explanations showing the effect of small compositional changes on predicted quality could help winemakers optimize ingredient modifications.

In addition, making AI visual dashboards where users can view real-time insights into feature importance and have interactions with those features would increase confidence and adoption.

E. Web Application Scalability and Deployment

For mass acceptance in the industry, the web-based AI system would need development that would support a wide range of concurrent users by an enterprise-sized wine production facility. Future development opportunities could include:

- Distributed Cloud-Based AI Inference for quondam prediction.
- Integration with existing wine inventory management systems so that firms will be able to adjust their stock to optimize quality trends.
- Mobile application development: In this application, winemakers and sommeliers will have access to predictions on the go.

There are opportunities involved with the deployment of cloud-based AI pipelines with API availability that will allow wineries and retail businesses to seamlessly integrate wine quality prediction into their operations.

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F. Ethical Considerations and AI Bias Reduction

Considerations regarding fair and close-to-bias predictions by artificial intelligence systems would have to be made as such systems become increasingly entrenched in the industry.

- Bias in data due to the favoritism towards certain regions or varieties may cause misdiagnosing, forecasting, or limitations of small-scale winemaking.
- There should be ethical AI frameworks in place to ensure that fair assessments are made for ill-resourced and nascent winemakers.
- Preventing the manipulation of wine scoring and market pricing must be balanced against transparency of the models.

Henceforth, changes in instances of fair AI models suggesting measures against potential biases in wine ratings must be put on the agenda for further research in order to secure both equal opportunity and equal assurance for all types of wine producers.

VII. Conclusion

This research has effectively developed a machine learning and Explainable AI (XAI)-driven web-based wine quality prediction system to render assessments driven by AI transparent and interpretable. Comparative modeling approaches were used to compare Random Forest with XGBoost and Artificial Neural Networks (ANN), establishing XGBoost as the most accurate, while Random Forest emerged as the most competent solution in balancing performance and interpretability. With SHAP and LIME to conduct a deep-dive into feature importance, it was determined that alcohol content, volatile acidity, and sulphates were the top three features affecting wine quality. The web application deployment means winemakers, retailers, and the end consumer can access AI-driven quality predictions, enhancing decision-making across the wine industry. To add, this investigation emphasizes moral AI's importance in dataset extension and integration with IoT for real-time monitoring to uplift predictive accuracy and industry adoption. However, future work should work on boosting deep learning models by incorporating sensory and expert review data alongside the explainability enhancements to create a more holistic and trustworthy AI for winemaking. The results of this study lay bare XAI-powered AI systems that have the potential to transform wine quality assessment so that predictions become more accurate, interpretable, and accessible for the global wine industry.

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